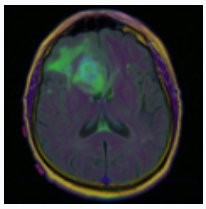
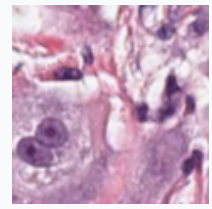
Stage 1: Multi-Modal Clinical Cohorts & Preprocessing


TCGA (MRI)


PANDA (Biopsy)


SIIM (X-Ray)


PanNuke (Tissue)

Patient-Isolated Partitioning | **Stain Normalization (Macenko)**

Input Image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$

Stage 2: Shared Backbone Encoder (VGG16 / MobileNetV2)

Block 1 128^2 → **Block 2** 64^2 → **Block 3** 32^2 → **Block 4** 16^2

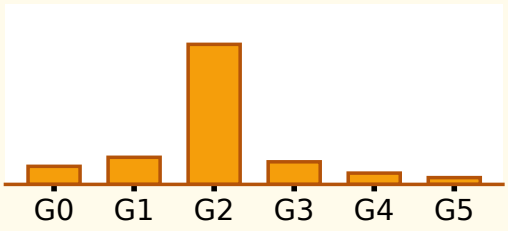
Lateral Skips $\{\mathbf{s}_l\}_{l=1}^{4}$

Shared Bottleneck Representation $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$
Lowest-Resolution Deep Feature Encoding (8×8 Latent Grid)

Bottleneck z | **Bottleneck z** | **Skips** $\{\mathbf{s}_l\}$

Stage 3: Task 1 (Classification)

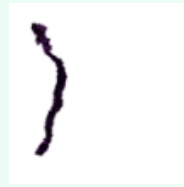
$\text{GAP} \rightarrow \text{MLP} \rightarrow \hat{\mathbf{y}}_{cls}$



Loss $\mathcal{L}_{cls}$ **(Categorical CE)**

Stage 3: Task 2 (Segmentation)

$\text{Dec}_\phi(\mathbf{z}, \{\mathbf{s}_l\}) \rightarrow \mathbf{I}_{seg}$



Loss $\mathcal{L}_{seg}$ **(Mask CE)**

Stage 4: Multi-Task Optimization & Evaluation Protocols

Multi-Task Optimization Dynamics
Static Loss Weighting ($\lambda_{cls} : \lambda_{seg}$) | Adaptive GradNorm Balancing

Clinical Evaluation Protocol
Disaggregated Foreground Dice (d_{fg}) | Empty-Credit Score Floor Guard (ρ)